

Continuum Mechanics

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Chapter 05

Balance Laws
Part 2



Local Conservation of Linear Momentum

Result 5.7 (Local Linear Momentum Balance)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion with spatial velocity $\mathbf{v}$ and mass density ρ. Axiom 5.2, balance of linear momentum, is equivalent to

$$\rho \dot{\mathbf{v}} = \nabla^{\mathrm{x}} \cdot S + \rho \mathbf{b}, \qquad \mathbf{x} \in B_t, \quad t > 0,$$

where $S(\mathbf{x}, t)$ is the Cauchy stress field and $\mathbf{b}(\mathbf{x}, t)$ is the spatial body force per unit mass.



Proof.

($\Rightarrow$): Assume the linear-momentum part of Axiom 5.2 holds. Then

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho\mathbf{v}\,\mathrm{d}V_{\mathrm{x}} &= \int_{\partial\Omega_t}\mathbf{t}\,\mathrm{d}A_{\mathrm{x}}+\int_{\Omega_t}\rho\mathbf{b}\,\mathrm{d}V_{\mathrm{x}}\\ &= \int_{\partial\Omega_t}S\mathbf{n}\,\mathrm{d}A_{\mathrm{x}}+\int_{\Omega_t}\rho\mathbf{b}\,\mathrm{d}V_{\mathrm{x}}\\ &= \int_{\Omega_t}(\nabla^{\mathrm{x}}\cdot S+\rho\mathbf{b})\,\mathrm{d}V_{\mathrm{x}},\end{aligned}$$

using the Divergence Theorem and $\mathbf{t}=S\mathbf{n}$ on $\partial\Omega_t$.

Proof (Cont.)

Using Result 5.6 componentwise,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho\mathbf{v}\;\mathrm{d}V_{\mathrm{x}}=\int_{\Omega_t}\rho\dot{\mathbf{v}}\;\mathrm{d}V_{\mathrm{x}}.$$

Therefore

$$\int_{\Omega_t}\left(\rho\dot{\mathbf{v}}-\nabla^{\mathrm{x}}\cdot S-\rho\mathbf{b}\right)\mathrm{d}V_{\mathrm{x}}=\mathbf{0}.$$

Since $\Omega_t \subseteq B_t$ is arbitrary, the Localization Theorem gives

$$\rho\dot{\mathbf{v}}=\nabla^{\mathrm{x}}\cdot S+\rho\mathbf{b}.$$

($\Leftarrow$): Integrate this local balance over Ω_t, then reverse the preceding steps. □



Local Balance of Angular Momentum

Result 5.8 (Local Angular Momentum Balance)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion and let $S : B_t \times [0, \infty) \to \mathcal{V}^2$ denote the Cauchy stress field in B_t. Then the balance of angular momentum requires

$$S^{\mathsf{T}} = S, \qquad \mathbf{x} \in B_t, \quad t > 0.$$

Proof.

From the balance of angular momentum, Axiom 5.2, for any $\Omega_t \subseteq B_t$,

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_t} \mathbf{x} \times (\rho \mathbf{v}) \, \mathrm{d}V_{\mathrm{x}} = \int_{\partial\Omega_t} \mathbf{x} \times \mathbf{t} \, \mathrm{d}A_{\mathrm{x}} + \int_{\Omega_t} \mathbf{x} \times (\rho \mathbf{b}) \, \mathrm{d}V_{\mathrm{x}}.$$

Proof (Cont.)

By Result 5.6, applied componentwise,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho(\mathbf{x},t)\big(\mathbf{x}\times\mathbf{v}(\mathbf{x},t)\big)\,\mathrm{d}V_{\mathrm{x}}=\int_{\Omega_t}\rho(\mathbf{x},t)\big(\mathbf{x}\times\mathbf{v}(\mathbf{x},t)\big)^{\cdot}\,\mathrm{d}V_{\mathrm{x}}.$$

Set

$$\mathbf{w}(\mathbf{x},t):=\mathbf{x}\times\mathbf{v}(\mathbf{x},t).$$

Then Result 4.7 gives

$$\dot{\mathbf{w}}(\mathbf{x},t)=\frac{\partial\mathbf{w}}{\partial t}(\mathbf{x},t)+\big(\nabla^{\mathrm{x}}\mathbf{w}(\mathbf{x},t)\big)\mathbf{v}(\mathbf{x},t).$$

Proof (Cont.)

Since $[\mathbf{x} \times \mathbf{v}]_i = \epsilon_{ik\ell} x_k v_\ell$,

$$
\begin{aligned}
[\nabla^{\mathrm{x}}(\mathbf{x} \times \mathbf{v})]_{ij} &= \frac{\partial}{\partial x_j}\left(\epsilon_{ik\ell} x_k v_\ell\right) \\
&= \epsilon_{ik\ell}\left(\frac{\partial x_k}{\partial x_j} v_\ell + x_k \frac{\partial v_\ell}{\partial x_j}\right) \\
&= \epsilon_{ik\ell}\delta_{kj} v_\ell + \epsilon_{ik\ell} x_k \frac{\partial v_\ell}{\partial x_j} \\
&= \epsilon_{ij\ell} v_\ell + \epsilon_{ik\ell} x_k \frac{\partial v_\ell}{\partial x_j}.
\end{aligned}
$$

Here i, j are free, while k, ℓ are dummy indices.

Proof (Cont.)

Therefore,

$$
\begin{aligned}
[\nabla^{\mathrm{x}}(\mathbf{x}\times\mathbf{v})\mathbf{v}]_i &= \epsilon_{ij\ell}v_\ell v_j + \epsilon_{ik\ell}x_k\frac{\partial v_\ell}{\partial x_j}v_j \\
&= \epsilon_{ik\ell}x_k\frac{\partial v_\ell}{\partial x_j}v_j \\
&= \left[\mathbf{x}\times\left((\nabla^{\mathrm{x}}\mathbf{v})\mathbf{v}\right)\right]_i .
\end{aligned}
$$

The first term vanishes because $v_\ell v_j$ is symmetric in j,ℓ, whereas $\epsilon_{ij\ell}$ is skew-symmetric in those indices.

Also, $\partial\mathbf{x}/\partial t = \mathbf{0}$ in the spatial description, so

$$
\frac{\partial}{\partial t}(\mathbf{x}\times\mathbf{v}) = \mathbf{x}\times\frac{\partial\mathbf{v}}{\partial t}.
$$

Proof (Cont.)

Combining the preceding identities,

$$\begin{aligned}\dot{\mathbf{w}}(\mathbf{x}, t) &= \mathbf{x} \times \left(\frac{\partial \mathbf{v}}{\partial t} + (\nabla^{\mathrm{x}} \mathbf{v})\mathbf{v} \right) \\ &= \mathbf{x} \times \dot{\mathbf{v}}.\end{aligned}$$

Consequently,

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_t} \rho(\mathbf{x} \times \mathbf{v}) \, \mathrm{d}V_{\mathrm{x}} = \int_{\Omega_t} \rho(\mathbf{x} \times \dot{\mathbf{v}}) \, \mathrm{d}V_{\mathrm{x}}.$$

The angular-momentum balance therefore becomes

$$\int_{\Omega_t} \rho(\mathbf{x} \times \dot{\mathbf{v}}) \, \mathrm{d}V_{\mathrm{x}} = \int_{\partial\Omega_t} \mathbf{x} \times \mathbf{t} \, \mathrm{d}A_{\mathrm{x}} + \int_{\Omega_t} \mathbf{x} \times (\rho \mathbf{b}) \, \mathrm{d}V_{\mathrm{x}}.$$

Proof (Cont.)

Recall that

$$\mathbf{t}(\mathbf{x}, t) = S(\mathbf{x}, t)\mathbf{n}(\mathbf{x})$$

and, by Result 5.7,

$$\rho\dot{\mathbf{v}} = \nabla^{\mathrm{x}} \cdot S + \rho\mathbf{b}.$$

Substituting both identities into the angular-momentum balance gives

$$\int_{\Omega_t} \mathbf{x} \times \left(\nabla^{\mathrm{x}} \cdot S + \rho\mathbf{b}\right) \, \mathrm{d}V_{\mathrm{x}} = \int_{\partial\Omega_t} \mathbf{x} \times (S\mathbf{n}) \, \mathrm{d}A_{\mathrm{x}} + \int_{\Omega_t} \mathbf{x} \times (\rho\mathbf{b}) \, \mathrm{d}V_{\mathrm{x}}.$$

Cancelling the body-force terms yields

$$\int_{\Omega_t} \mathbf{x} \times (\nabla^{\mathrm{x}} \cdot S) \, \mathrm{d}V_{\mathrm{x}} = \int_{\partial\Omega_t} \mathbf{x} \times (S\mathbf{n}) \, \mathrm{d}A_{\mathrm{x}}.$$

Proof (Cont.)

From this point, the calculation is the same as the proof of Result 3.5; specifically, see equation (3.9). That argument gives

$$\int_{\partial\Omega_t} \mathbf{x} \times (S\mathbf{n}) \, \mathrm{d}A_{\mathrm{x}} - \int_{\Omega_t} \mathbf{x} \times (\nabla^{\mathrm{x}} \cdot S) \, \mathrm{d}V_{\mathrm{x}} = \int_{\Omega_t} \boldsymbol{\epsilon} : S \, \mathrm{d}V_{\mathrm{x}},$$

where $[\boldsymbol{\epsilon} : S]_i = \epsilon_{ijk} S_{kj}$.

Thus $\boldsymbol{\epsilon} : S = \mathbf{0}$ by localization. Equation (3.9), or its componentwise case calculation, then implies

$$S_{ij} = S_{ji}, \qquad \text{hence} \qquad S^{\mathsf{T}} = S.$$





Local Eulerian Version of the First Law

Result 5.9 (Net Working)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of a continuum body associated with the spatial velocity field $\mathbf{v}$ and Cauchy stress tensor S. Let

$$L(\mathbf{x}, t) = \frac{1}{2}\left(\nabla^{\mathrm{x}}\mathbf{v} + (\nabla^{\mathrm{x}}\mathbf{v})^{\mathsf{T}}\right) = \mathsf{sym}(\nabla^{\mathrm{x}}\mathbf{v})$$

denote the rate-of-strain field. For arbitrary $\Omega_t \subseteq B_t$,

$$\frac{\mathrm{d}}{\mathrm{d}t}K[\Omega_t] + \int_{\Omega_t} S : L \, \mathrm{d}V_{\mathrm{x}} = \mathcal{P}[\Omega_t], \qquad t > 0.$$

Thus, in view of (5.14),

$$\mathcal{W}[\Omega_t] = \int_{\Omega_t} S : L \, \mathrm{d}V_{\mathrm{x}}.$$

Proof.

By conservation of linear momentum,

$$\rho\dot{\mathbf{v}} = \nabla^{\mathrm{x}} \cdot S + \rho\mathbf{b}.$$

Taking the dot product with $\mathbf{v}$ gives

$$\rho\dot{\mathbf{v}} \cdot \mathbf{v} = (\nabla^{\mathrm{x}} \cdot S) \cdot \mathbf{v} + \rho\mathbf{b} \cdot \mathbf{v}.$$

Result 2.11 states

$$\nabla^{\mathrm{x}} \cdot (S^{\mathsf{T}}\mathbf{v}) = (\nabla^{\mathrm{x}} \cdot S) \cdot \mathbf{v} + S : \nabla^{\mathrm{x}}\mathbf{v}.$$

Therefore,

$$\rho\dot{\mathbf{v}} \cdot \mathbf{v} = \nabla^{\mathrm{x}} \cdot (S^{\mathsf{T}}\mathbf{v}) - S : \nabla^{\mathrm{x}}\mathbf{v} + \rho\mathbf{b} \cdot \mathbf{v}.$$

Proof (Cont.)

Integrating over Ω_t and using the Divergence Theorem,

$$\begin{aligned}\int_{\Omega_t} \rho\dot{\mathbf{v}}\cdot\mathbf{v}\,\mathrm{d}V_{\mathrm{x}} &= \int_{\Omega_t} \nabla^{\mathrm{x}}\cdot(S^{\mathsf{T}}\mathbf{v})\,\mathrm{d}V_{\mathrm{x}} - \int_{\Omega_t} S:\nabla^{\mathrm{x}}\mathbf{v}\,\mathrm{d}V_{\mathrm{x}} \\ &\quad + \int_{\Omega_t} \rho\mathbf{b}\cdot\mathbf{v}\,\mathrm{d}V_{\mathrm{x}} \\ &= \int_{\partial\Omega_t} (S^{\mathsf{T}}\mathbf{v})\cdot\mathbf{n}\,\mathrm{d}A_{\mathrm{x}} - \int_{\Omega_t} S:\nabla^{\mathrm{x}}\mathbf{v}\,\mathrm{d}V_{\mathrm{x}} \\ &\quad + \int_{\Omega_t} \rho\mathbf{b}\cdot\mathbf{v}\,\mathrm{d}V_{\mathrm{x}}.\end{aligned}$$

Proof (Cont.)

Since $S^{\mathsf{T}} = S$ and $\mathbf{t} = S\mathbf{n}$,

$$(S^{\mathsf{T}}\mathbf{v}) \cdot \mathbf{n} = \mathbf{v} \cdot (S\mathbf{n}) = \mathbf{v} \cdot \mathbf{t}.$$

Moreover, because S is symmetric,

$$S : \nabla^{\mathsf{x}}\mathbf{v} = S : \mathsf{sym}(\nabla^{\mathsf{x}}\mathbf{v}) = S : L.$$

Hence

$$\begin{aligned}\int_{\Omega_t} \rho\dot{\mathbf{v}} \cdot \mathbf{v} \, \mathrm{d}V_{\mathsf{x}} + \int_{\Omega_t} S : L \, \mathrm{d}V_{\mathsf{x}} &= \int_{\partial\Omega_t} \mathbf{v} \cdot \mathbf{t} \, \mathrm{d}A_{\mathsf{x}} \\ &\quad + \int_{\Omega_t} \rho\mathbf{b} \cdot \mathbf{v} \, \mathrm{d}V_{\mathsf{x}} \\ &= \mathcal{P}[\Omega_t].\end{aligned}$$

Proof (Cont.)

Finally, by Result 5.6,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}K[\Omega_t] &= \frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\frac{1}{2}\rho(\mathbf{v}\cdot\mathbf{v})\,\mathrm{d}V_{\mathrm{x}}\\ &= \frac{1}{2}\int_{\Omega_t}\rho(\mathbf{v}\cdot\mathbf{v})^{\cdot}\,\mathrm{d}V_{\mathrm{x}}\\ &= \int_{\Omega_t}\rho\dot{\mathbf{v}}\cdot\mathbf{v}\,\mathrm{d}V_{\mathrm{x}}.\end{aligned}$$

Substitution gives

$$\frac{\mathrm{d}}{\mathrm{d}t}K[\Omega_t] + \int_{\Omega_t} S : L\,\mathrm{d}V_{\mathrm{x}} = \mathcal{P}[\Omega_t].$$

Using $\mathcal{W} = \mathcal{P} - \frac{\mathrm{d}}{\mathrm{d}t}K$ proves the second statement. □



Result 5.10 (Local Eulerian First Law)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of a continuum body with spatial mass density $\rho(\mathbf{x}, t)$, associated velocity $\mathbf{v}$, and internal energy density per unit mass ϕ. Then the First Law of Thermodynamics implies

$$\rho \dot{\phi} = S : L - \nabla^{\mathrm{x}} \cdot \mathbf{q} + \rho r, \qquad \mathbf{x} \in B_t, \quad t > 0.$$



Proof.

The First Law of Thermodynamics states that

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho\phi\,\mathrm{d}V_x = Q[\Omega_t] + \mathcal{W}[\Omega_t].$$

Using Result 5.9 and the definition of net heating,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho\phi\,\mathrm{d}V_x = \int_{\Omega_t}\rho r\,\mathrm{d}V_x - \int_{\partial\Omega_t}\mathbf{q}\cdot\mathbf{n}\,\mathrm{d}A_x + \int_{\Omega_t} S : L\,\mathrm{d}V_x.$$

By Result 5.6,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho\phi\,\mathrm{d}V_x = \int_{\Omega_t}\rho\dot{\phi}\,\mathrm{d}V_x.$$

Proof (Cont.)

The Divergence Theorem gives

$$\int_{\partial\Omega_t} \mathbf{q}\cdot\mathbf{n}\,\mathrm{d}A_{\mathrm{x}} = \int_{\Omega_t} \nabla^{\mathrm{x}}\cdot\mathbf{q}\,\mathrm{d}V_{\mathrm{x}}.$$

Thus, for any $\Omega_t \subseteq B_t$,

$$\int_{\Omega_t} \left\{\rho\dot{\phi} + \nabla^{\mathrm{x}}\cdot\mathbf{q} - \rho r - S : L\right\}\mathrm{d}V_{\mathrm{x}} = 0.$$

By the Localization Theorem, since Ω_t is arbitrary,

$$\rho\dot{\phi} = S : L - \nabla^{\mathrm{x}}\cdot\mathbf{q} + \rho r, \qquad \mathbf{x}\in B_t, \quad t > 0.$$



Local Eulerian Version of the Second Law

Result 5.11 (Local Eulerian Second Law)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of a continuum body with mass density $\rho(\mathbf{x}, t)$ and entropy density per unit mass $\eta(\mathbf{x}, t)$. Then the Second Law of Thermodynamics implies

$$\rho \dot{\eta} = \frac{\rho r}{\theta} - \nabla^{\mathrm{x}} \cdot \left(\frac{\mathbf{q}}{\theta} \right) + \sigma, \qquad \mathbf{x} \in B_t, \quad t > 0,$$

where the entropy production rate σ is everywhere and always nonnegative:

$$\sigma \geq 0, \qquad \mathbf{x} \in B_t, \quad t > 0.$$



Proof.

The Second Law, Axiom 5.4, states

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho\eta\,\mathrm{d}V_{\mathrm{x}}-\left[\int_{\Omega_t}\frac{\rho r}{\theta}\,\mathrm{d}V_{\mathrm{x}}-\int_{\partial\Omega_t}\frac{\mathbf{q}\cdot\mathbf{n}}{\theta}\,\mathrm{d}A_{\mathrm{x}}\right]=\int_{\Omega_t}\sigma\,\mathrm{d}V_{\mathrm{x}}.$$

Equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho\eta\,\mathrm{d}V_{\mathrm{x}}=\int_{\Omega_t}\frac{\rho r}{\theta}\,\mathrm{d}V_{\mathrm{x}}-\int_{\partial\Omega_t}\left(\frac{\mathbf{q}}{\theta}\right)\cdot\mathbf{n}\,\mathrm{d}A_{\mathrm{x}}+\int_{\Omega_t}\sigma\,\mathrm{d}V_{\mathrm{x}}.$$

Proof (Cont.)

By Result 5.6,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho\eta\,\mathrm{d}V_{\mathrm{x}}=\int_{\Omega_t}\rho\dot{\eta}\,\mathrm{d}V_{\mathrm{x}}.$$

By the Divergence Theorem,

$$\int_{\partial\Omega_t}\left(\frac{\mathbf{q}}{\theta}\right)\cdot\mathbf{n}\,\mathrm{d}A_{\mathrm{x}}=\int_{\Omega_t}\nabla^{\mathrm{x}}\cdot\left(\frac{\mathbf{q}}{\theta}\right)\mathrm{d}V_{\mathrm{x}}.$$

Thus

$$\int_{\Omega_t}\left\{\rho\dot{\eta}-\frac{\rho r}{\theta}+\nabla^{\mathrm{x}}\cdot\left(\frac{\mathbf{q}}{\theta}\right)-\sigma\right\}\mathrm{d}V_{\mathrm{x}}=0.$$

Proof (Cont.)

Since $\Omega_t \subseteq B_t$ is arbitrary, the Localization Theorem gives

$$\rho\dot{\eta} - \frac{\rho r}{\theta} + \nabla^{\mathrm{x}} \cdot \left(\frac{\mathbf{q}}{\theta}\right) - \sigma = 0.$$

Rearranging,

$$\rho\dot{\eta} = \frac{\rho r}{\theta} - \nabla^{\mathrm{x}} \cdot \left(\frac{\mathbf{q}}{\theta}\right) + \sigma, \qquad \mathbf{x} \in B_t, \quad t > 0.$$

Finally, Axiom 5.4 requires

$$\int_{\Omega_t} \sigma \, \mathrm{d}V_{\mathrm{x}} \geq 0 \qquad \text{for every } \Omega_t \subseteq B_t.$$

Localization therefore yields

$$\sigma(\mathbf{x}, t) \geq 0, \qquad \mathbf{x} \in B_t, \quad t > 0.$$

□